Do VLMs Understand Aggression? A Benchmark and Progressive Fine-Tuning Pipeline for Video Aggression Detection

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Abstract

Vision-language models (VLMs) are increasingly deployed for video understanding, yet their ability to reason DY: incorrect grammar, "about" shouldn't be here about *who* did *what* to *whom* in aggressive interactions remains largely unexamined. We introduce a benchmark of 2,683 video clips spanning 17 aggression categories, accompanied by over 15K multiple-choice questions organized into three tiers of compositional difficulty, from single-attribute identification to full aggressor-action-victim reasoning. Each question's distractors are constructed from a 10-level hardness taxonomy that systematically controls for shortcut cues such as role reversal, bystander substitution, and cross-video borrowing. Zero-shot evaluation of 13 MLLMs reveals that most model achieves only 50% accuracy, with performance dropping further below on detailed questions requiring joint aggressor-action-victim understanding.

To address this gap, we propose a progressive fine-tuning pipeline: supervised fine-tuning teaches MCQ format compliance, chain-of-thought distillation adds structured reasoning for compositional questions, and anchored direct preference optimization sharpens discrimination against taxonomy-hard distractors. Training on just 20% of videos with group-aware splits that prevent parent-video leakage, our pipeline improves accuracy to **XX%** (+XX points), with the largest gains on the compositional categories that zero-shot models struggle with most.

1 Introduction

Recent Multimodal Large Language Models (MLLMs) have demonstrated strong performance on video understanding tasks, including action recognition, temporal event localization, and video question answering. Existing action datasets [10, 15, 36, 22, 19] have been highly effective in evaluating models across a wide range of everyday activities, sports actions, human-object interactions, and complex temporal events. While these tasks are important for general video understanding, reliable real-world systems must also recognize safety-critical behaviors, including physical altercations or threatening interactions. However, understanding aggression requires more than detecting that a violent or harmful action has occurred. In practical settings such as surveillance, public safety monitoring, school safety, sports analysis, and online video moderation, a model must also identify how the people in the scene are involved. Without role-aware reasoning, a system may correctly classify a scene as aggressive while still producing an incorrect or harmful interpretation.

Prior work has studied aggressive actions through crime detection, violence detection, and incident understanding datasets. [37, 44, 5, 4, 6] provide multiple aggressive action categories, while others [12, 21, 27, 29, 31, 18] formulate the task as binary classification, such as normal vs. anomalous or violent vs. non-violent. These resources have advanced the ability of models to detect abnormal or

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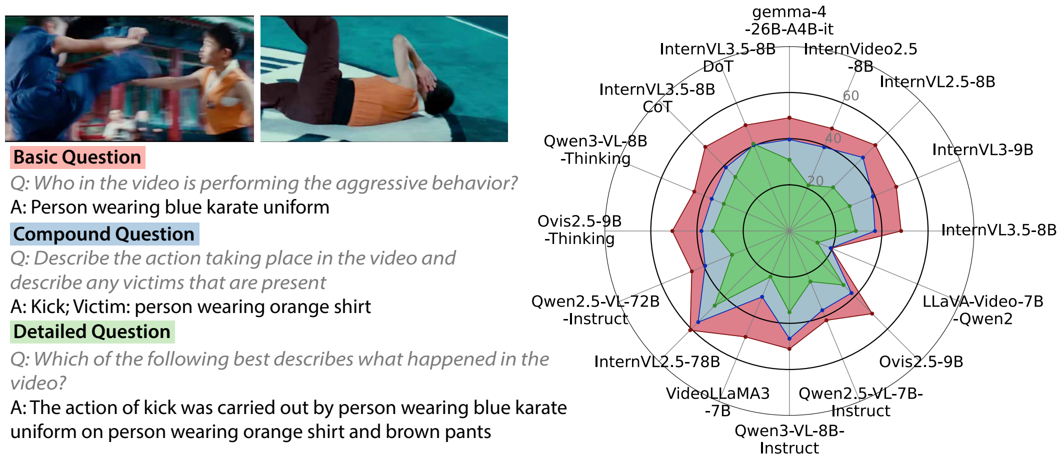


Figure 1: **A^{3}Bench evaluates fine-grained aggressive-action association.** *Left:* Example question types from A^{3}Bench. The benchmark progresses from basic questions that ask about a single attribute, to detailed questions that require understanding all aspects. *Right:* Performance of evaluated MLLMs across the three question levels. Models perform poorly overall most scoring below 50%, and accuracy decreases as the task becomes more difficult, with the lowest scores on detailed questions. This highlights the difficulty of moving from coarse aggression recognition to reliable actor-role and action binding.

dangerous events, but they typically evaluate only the existence of aggression. They often do not require models to reason deeply about the individuals involved in the scene. In particular, existing evaluations rarely test whether a model can bind an aggressive action to the correct actor, distinguish aggressors from victims and bystanders, or resolve multi-person interactions. This limitation can reward models that rely on contextual shortcuts, scene priors, or coarse motion cues, rather than learning the semantic relationships between people, actions, and roles. This necessitates a more fine-grained capability: **aggressive-action association**, where the goal is not only to recognize an aggressive behavior, but also to associate it with the correct participant roles.

To address this, we introduce A^{3}Bench, a benchmark for evaluating Aggressive-Action Association in MLLMs. A^{3}Bench contains 2,683 video clips and 15,004 video question-answer pairs designed to probe models beyond coarse aggression recognition. Instead of asking only whether a violent action occurs, our benchmark asks detailed questions about the type of aggressive action, and action-role relationships (aggressors, victims, bystanders). These questions require models to interpret the scene composition, distinguish the appearance between individuals, and bind actions to the correct participants. By evaluating these fine-grained reasoning abilities, A^{3}Bench provides a focused testbed for assessing whether current MLLMs can reliably understand aggressive scenes at the level required for real-world deployment. Our experiments show that recent MLLMs struggle with this setting with most models achieving performance below 50%. RA: Talk about solution?

Overall, our contributions are as follows:

- We introduce A^{3}Bench, a benchmark of 2,683 video clips designed to evaluate fine-grained aggressive behavior understanding in MLLMs.
- We construct diverse question types of 12,773 video question-answer pairs for deep understanding by probing to role identification along with aggressive action understanding.
- We benchmark recent MLLMs revealing limitations in their ability to properly understand aggressive scenes.
- RA: Solution?

Table 1: **Comparison of A^{3}Bench with prior crime/violence/aggressive action benchmarks.** Role Definitions indicates whether explicit semantic roles are annotated for the subjects involved in the action. Person Description indicates whether natural-language descriptions are provided for people in the video.

Benchmark	# of Classes	Role Definition	Person Description	VQA Questions
DVD [21]	2	×	×	×
Movies Fight [31]	2	×	×	×
RLVS [9]	2	×	×	×
RWF-2000 [12]	2	×	×	×
VID [29]	2	×	×	×
Violent Flows [9]	2	×	×	×
CABAD [5]	6	×	×	×
CRxK [6]	13	×	×	×
IARD [4]	4	×	×	×
UCF-Crime [37]	14	×	×	×
XD-Violence [44]	7	×	×	×
A^{3}Bench (Ours)	**18**	✓	✓	✓

2 Related Work

Action recognition: The advancement of action recognition can be noted across video [38, 39], skeleton [45, 51, 41], and multimodal modalities [34, 40, 30, 20, 1, 13]. Certain methods, such as temporal localization [35, 24, 25] allow more precise event boundary detection under low-cost data annotation. Fine-grained recognition has been moved towards by unified keypoint-based frameworks [17] and atomic action decomposition [32]. However, these methods treat action recognition as a classification problem over participant-agnostic categories without modeling the relationship between those involved, a distinction critical to have practical usage for the data.

Crime and anomaly detection: Anomaly detection and behavior analysis in surveillance videos have been widely studied for identifying abnormal, unsafe, or criminal activities [49, 23]. In parallel, dedicated violence and aggression datasets have advanced the evaluation of models across diverse harmful scenarios, including fight detection, crowd violence, real-world surveillance violence, intruder activity, child aggression, and crime/anomaly recognition [31, 21, 9, 18, 12, 29, 4, 44, 5, 6, 37]. Beyond visual understanding, text-based aggression detection has also examined aggressive behavior in language, such as harmful or hostile social-media comments [2], showing that aggression analysis spans multiple modalities. However, most existing video benchmarks primarily focus on detecting whether violence, aggression, or anomaly is present, or on assigning coarse event-level labels.

Multimodal Large Language Models (MLLMs): Recent multimodal models have significantly expanded the scope of visual understanding by linking visual evidence with language-based semantic reasoning [33]. Building on large-scale cross-modal pretraining, such models have enabled zero-shot and open-vocabulary recognition of human actions and activities [3, 40, 30, 34, 20, 1]. Recent video-centric MLLMs [7, 52, 26, 14], further extend these capabilities to temporal and multi-frame reasoning over complex events [28, 46, 50]. These advances motivate the use of MLLMs for aggressive action and behavior understanding, where models must not only identify the observed action but also reason about the relationships between people and their roles within the interaction.

RA: Prior solution methods (Will write this later if solution works)

3 Benchmarking Aggressive-Action Association

3.1 Construction

Video curation and annotation: We curate a benchmark of 2,683 video clips from diverse sources, including publicly available social-media videos, UCF-Crime [37], and RWF-2000 [12]. The videos span a wide range of recording conditions, camera viewpoints, resolutions, scene layouts, and levels

Figure 2: **Example videos from A^{3}Bench.** Each video is annotated with structured role and scene information, including the aggressor, victim, bystanders, aggressive action, and environment. Person descriptions are based primarily on clothing and coarse visual attributes rather than identity, enabling fine-grained actor-role association while avoiding personally identifying information.

of visual clutter, reflecting the heterogeneity of real-world footage. Long videos are trimmed into shorter clips with an average duration of 5.75 seconds. The final corpus contains 2,234 clips with aggressive behavior and 449 non-aggressive clips.

A group of annotators labels each video with structured scene information. The annotations include: (1) the aggressive action being performed, selected from 17 aggressive action categories and a control "none" category for non-aggressive clips; a short appearance description of the (2) aggressor; (3) victim; (4) bystanders (if any) and (5) the environment or setting. As seen in Figure 2, person descriptions are primarily based on clothing and coarse visual appearance, such as "person in a dark jacket," while avoiding names, faces, demographic attributes, or other identifying information. This design reduces the risk of relying on sensitive attributes and focuses the benchmark on visual role and action understanding. If a certain role does not exist in the video clip, or the appearance is too ambiguous, that is left out in the annotations. More details are provided in Figure 3.

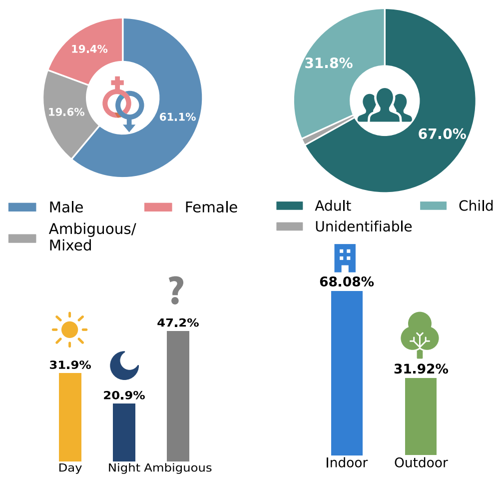
Question generation and distractor design: From the annotation corpus, we generate multiple-choice video question-answer pairs designed to probe different aspects of aggressive behavior understanding. The questions are organized into three tiers of increasing complexity. Figure 1 (left) provides an overview of question types, while Figure 4 shows total distribution.

Basic questions test individual attributes of the scene. These include primary action identification, where the model must recognize the aggressive action taking place; aggressor/victim identification, where the model must identify who performs the aggression or who is the target; and role identification, where the model is given a person description and must assign the correct role, such as aggressor, victim, or bystander.

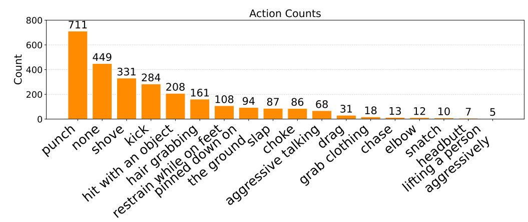
Compound questions require jointly identifying two attributes such as: aggressor-victim, action-aggressor and action-victim association. These questions require selecting the choice where all components of the answer are correct. This design directly evaluates whether models can bind actions and roles to the appropriate participants and scene context.

Detailed questions require holistic understanding of the interaction. These include aggressor-action-victim questions, which ask who did what to whom, and sequence verification questions, which require selecting the correct structured description of the full interaction. These test whether a model can reason about the directionality of aggression, distinguish participants with different roles, and construct a coherent interpretation of the event. In total, the benchmark contains 15,004 questions, corresponding to approximately 5.6 questions per video.

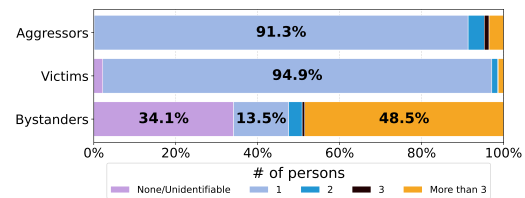
We design distractor choices to be plausible and semantically challenging.



(a) Demography and environment settings



(b) Action counts



(c) Distribution of annotated roles per video

Figure 3: **Dataset statistics for A^{3}Bench.** (a) A^{3}Bench contains a larger proportion of male and adult participants among videos where demographic attributes are identifiable. For scene conditions, many clips are ambiguous in time of day, but among identifiable cases, slightly more aggressive clips occur during daytime than nighttime, and most are captured in indoor environments. (b) The action distribution is long-tailed. (c) The role-count distribution is also imbalanced: most clips contain a single identifiable aggressor and victim, while bystanders are more variable and often appear in larger groups.

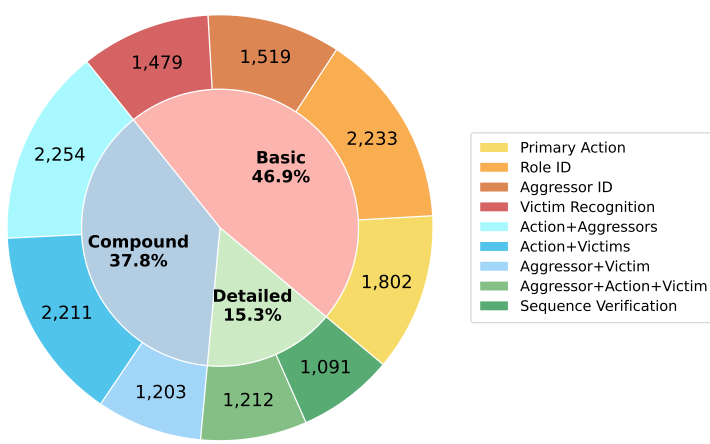


Figure 4: **Distribution of question types in A^{3}Bench.** The benchmark contains three levels of questions with increasing association complexity: *Basic* questions focus on single attributes; *Compound* questions require associating two scene components; and *Detailed* questions require full interaction understanding.

1. **Cross-video distractors.** Incorrect answers are drawn from real annotations of other videos in the dataset, ensuring that distractors are natural and visually plausible.
2. **Same-video distractors.** For person identification questions, we sample individuals of other roles visible in the same video as distractors. For example, when asking for the aggressor, the victim or a bystander may appear as an incorrect option. The model cannot trivially eliminate these choices because the person is presented in the video.
3. **Role reversals.** For compound and sequence questions, we construct distractors by swapping the aggressor and victim roles. This tests whether models understand the directionality of the interaction rather than merely detecting the presence of the correct people or action. We

also place bystanders in active roles to test whether models can distinguish participation from mere presence.

4. **Trick questions.** Some questions use negation as the correct answer, such as "no aggressive action is taking place." These questions prevent models from assuming that aggression is always present and require them to verify the visual content of the clip.

A key concern with multiple-choice is whether models exploit statistical regularities in the answer text rather than grounding responses in the video. To counteract frequency shortcuts in Detailed questions, we employ frequency-inverted distractor construction where each distractor property appears in multiple options, making marginal frequency uninformative.

Quality assurance: We apply several quality-control steps to ensure that the benchmark evaluates aggressive-action association rather than annotation artifacts. After initial video collection, clips with unclear aggression, confusing role assignments, or unresolved ambiguity between participants were removed from the final benchmark. All retained clips and annotations are manually cross-checked by multiple annotators. For question generation, we perform semantic similarity filtering to remove near-duplicate answer choices that could otherwise provide unintended cues. We exclude clips, or avoid generating questions for clips, in which the action, or person appearance is too ambiguous for reliable annotation, as well as clips where multiple people have highly similar clothing or appearance. When more than three individuals share the same role, we annotate them collectively and avoid generating fine-grained person-level questions that would be visually underdetermined.

3.2 Benchmarked Models

We evaluate **13** representative MLLM configurations on **A^{3}Bench** to assess their ability to perform aggressive-action association. All of our chosen models are trained on very large datasets with high-level video understanding.

General-Purpose Models. The evaluated models cover several recent open-weight MLLM families, including Gemma [14], InternVideo [43], InternVL [11, 52, 42], LLaVA-Video [48], Ovis [26], Qwen-VL [8, 7], and VideoLLaMA [47]. These families differ in visual encoder design, video modeling strategy, instruction tuning, model scale, and reasoning behavior, providing a broad comparison of current MLLMs on fine-grained aggressive scene understanding.

Reasoning and Prompted Variants. To investigate whether explicit reasoning improves aggressive-action association, we additionally evaluate reasoning-oriented variant of Qwen-VL-8B and enable reasoning-mode in Ovis2.5-9B. We also evaluate InternVL3.5-8B under two reasoning-style prompting settings: a detailed chain-of-thought (CoT) prompt and a Dream-of-Thoughts (inspired from [16]) prompt. These variants encourage intermediate reasoning before producing a final answer, allowing us to test whether step-by-step inference helps models distinguish aggressors, victims, bystanders, actions, and scene context.

3.3 Implementation Details, Evaluation Protocol, and Metrics

All models are evaluated using their official implementations and released weights. The 70B+ parameter models were run on their quantized versions to fit in GPU. We use a multiple-choice video question answering (VQA) protocol across all models: for each video-question pair, the model is prompted to select the most appropriate answer from the provided options. Most question types contain eight answer choices, consisting of one correct answer and seven distractors, while role identification questions contain four choices due to the constrained role label space. The position of the correct answer is uniformly shuffled across all questions to mitigate positional bias. This controlled evaluation setting enables fair comparison across models while directly measuring each model's ability to associate aggressive actions with the correct participants and roles. We report accuracy as the primary evaluation metric.

4 Results

Table 2 summarizes model performance on A^{3}Bench. Overall, current MLLMs perform poorly on the benchmark. Across all evaluated models, the average overall accuracy is only 41.59%, and nearly all models remain below 50%. Even the strongest model reaches only 56.60%, encountering difficulty

Table 2: **Benchmarking models on A^{3}Bench.** Reported metric is accuracy (%) for Basic Questions, Compound Questions, Detailed Questions, and Overall. **Best** and second-best performance are highlighted. RA: Teaser figure right has similar information. Will redundancy be an issue?

Model	Basic ↑	Compound ↑	Detailed ↑	Overall ↑
gemma-4-26B-A4B-it [14]	49.07	39.65	30.86	42.72
InternVideo2.5-8B [43]	48.08	39.27	21.42	40.66
InternVL2.5-8B [43]	52.61	45.01	26.75	45.77
InternVL3-9B [52]	50.05	39.06	28.22	42.55
InternVL3.5-8B [42]	48.34	37.03	28.75	41.06
LLaVA-Video-7B-Qwen2 [48]	19.58	19.34	13.11	18.50
Ovis2.5-9B [26]	50.62	37.99	33.09	43.16
Qwen2.5-VL-7B-Instruct [8]	41.87	37.15	23.73	37.30
Qwen3-VL-8B-Instruct [7]	51.07	46.65	35.19	46.96
VideoLLaMA3-7B [47]	49.65	30.86	21.41	38.22
InternVL2.5-78B-AWQ [43]	**60.76**	**55.84**	**45.77**	**56.60**
Qwen2.5-VL-72B-Instruct-AWQ [8]	45.73	39.47	26.91	40.50
Ovis2.5-9B-Thinking [26]	50.69	38.09	33.13	43.24
Qwen3-VL-8B-Thinking [7]	44.59	36.41	30.77	39.37
InternVL3.5-8B CoT prompt	51.53	38.92	33.13	43.94
InternVL3.5-8B DoT [16]	49.64	40.31	40.90	44.77

in aggressive-action association. This low performance suggests that current MLLMs are not yet reliable for fine-grained aggressive scene understanding, especially when evaluation goes beyond coarse recognition of whether aggression is present.

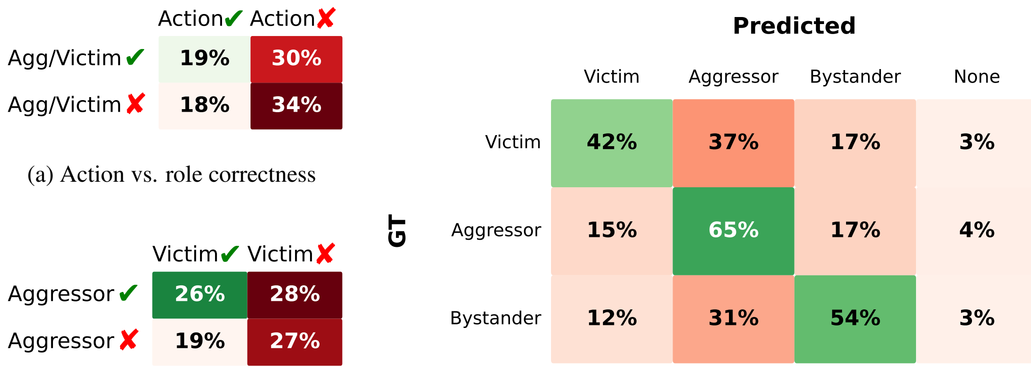
4.1 Analysis

Models struggle as questions require deeper understanding. Models achieve an average accuracy of 47.74% on *Basic* questions, but this drops to 38.82% on *Compound* questions and further to 29.58% on *Detailed* questions. This consistent decline indicates that models can sometimes recognize isolated cues, such as an action category or a visually salient person, but struggle when they must jointly reason about multiple scene components. In particular, *Detailed* questions require models to understand the full interaction structure, such as who did what to whom, exposing failures in directionality, role assignment, and person-action binding.

Explicit reasoning does not reliably improve association. Overall gains for reasoning-style variants are limited and they do not outperform the strongest general-purpose model. The average overall accuracy of the reasoning-style models remains only 42.83%, which is still below reliable. This suggests that simply encouraging models to produce intermediate reasoning is not sufficient to solve aggressive-action association, and that current MLLMs still lack robust mechanisms for grounding aggressive actions to the correct participants and roles.

Models struggle more with recognizing aggressive actions than basic role identification. To understand whether errors arise from recognizing the aggressive behavior or identifying the relevant participants, we analyze videos that include both a primary action question and at least one aggressor or victim identification question. Figure 5(a) reports results averaged across models. Notably, models are more likely to identify the correct aggressor or victim while failing to recognize the correct action (30%) than to recognize the correct action while failing to identify the relevant role (18%). This suggests that, even for basic questions, current MLLMs often struggle to distinguish fine-grained aggressive actions, and that person-level saliency alone does not ensure accurate understanding of what aggressive behavior is taking place.

Victim recognition remains a key failure mode. We analyze role-related errors using videos that include separate basic questions for both aggressor and victim identification, with results averaged across models in Figure 5(b). When only one role is correct, models are more likely to identify the aggressor while misidentifying the victim (28%) than the other way around (19%), suggesting that victim recognition is more challenging. This trend is reinforced by the confusion matrix for role



(a) Action vs. role correctness

(b) Aggressor vs. victim correctness

(c) Role identification confusion matrix

Figure 5: **Error analysis of action and role association.** We analyze model failures on basic questions by decomposing predictions across primary aggressive action recognition, aggressor/victim identification, and role classification. The results indicate that current MLLMs struggle not only to recognize aggressive actions, but also to assign the correct roles to the people involved.

identification questions in Figure 5(c): victims and bystanders are frequently predicted as aggressors, indicating a tendency to over-assign the aggressive role to visually salient or active individuals.

Social Appropriateness: We conduct a simple free-form evaluation on five models from the Ovis, InternVL, and Qwen families [26, 11, 7] which have relative better scores. Each model is asked to characterize the social appropriateness of the actions shown in a video. Models are generally able to identify aggressive or socially inappropriate behavior in the 2,238 aggressive clips. However, on the 449 non-aggressive clips, the models frequently still attribute inappropriate or aggressive actions to the videos, achieving below $\sim 60\%$ accuracy, which is close to random binary performance. This suggests that while current models are sensitive to aggressive cues, they also exhibit a bias toward over-predicting aggression, even when no aggressive behavior is present.

Distractors reveal role ambiguity and shortcut reasoning. Observing the same five models, errors are largely driven by distractors that are either visually grounded in the scene or statistically frequent in the answer choices. For Basic person-identification questions, same-video role-based distractors are more challenging than cross-video distractors: models are often confused when an incorrect option refers to another visible participant, especially in role-reversal distractors where the victim, aggressor, or bystander appears in the clip but is assigned the wrong role. For Detailed questions, models are often attracted to frequency-saturated distractors, where certain action-role-person combinations appear more frequently among the options. Instead of grounding predictions in visual evidence, models appear to rely on answer-choice priors, such as selecting a person description that frequently appears with a victim or aggressor role.

Understanding aggressive action scene with environment: We asked the model to identify the environment the action is happening in along with the action for a subset of videos where environment is clearly identifiable. Comparing them further shows that adding scene-grounding requirements makes the task harder for most models. On average, accuracy drops from 40.53% for primary action recognition to 36.69% for action-environment association. This suggests that models are not only challenged by recognizing aggressive actions, but also by grounding those actions in the correct environmental context. While identifying an action can often rely on local motion or appearance cues, action-environment questions require models to jointly reason about the behavior and the surrounding scene.

5 Solution

ASSUMING ADPO + SFT + COT

6 Conclusion

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    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.